

Panel MS-AR(1) after a common random-walk trend

Seasonally adjusted real GDP per worker

1 Model

Two-step. Step 1 builds a *common* trend from average growth of countries observed in consecutive quarters (not from the level mean, which falls when poorer countries enter). Log seasonally adjusted real GDP per worker, 2026-09-17 vintage, from 1987Q1 onward.

$$\Delta\tau_t = \frac{1}{n_t^{\text{cont}}} \sum_{i \in C_t \cap C_{t-1}} (y_{it} - y_{i,t-1}), \quad (1)$$

$$\tau_t = \tau_{t_0} + \sum_{s=t_0+1}^t \Delta\tau_s. \quad (2)$$

The level of τ is pinned to the balanced-core mean at t_0 . Mean $\Delta\tau$ is 0.0045 per quarter (3 countries present in every quarter). Step 2 is estimated on $y_{it}^* = y_{it} - \hat{\tau}_t$, with no extra *gt*:

$$y_{it}^* = a_i + b_i \lambda^{t-T_{i0}} + z_{it}, \quad (3)$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2), \quad b_i \sim \mathcal{N}(0, \omega_b^2) \text{ independent}, \quad (4)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it})z_{it} + \sigma(s_{it})\varepsilon_{it}. \quad (5)$$

$\lambda = 0.98$ is fixed (Barro 2% per year). $|\rho| < 0.99$. Three regimes; median μ pinned at 0; σ and ρ switch. T_{i0} is country i 's first observation after filters. Random effects: 7×7 Gauss–Hermite. Plotted cycles use posterior-mean a_i and b_i . The common trend τ_t is discarded for the structural MS-AR. Standard errors ignore step-1 uncertainty.

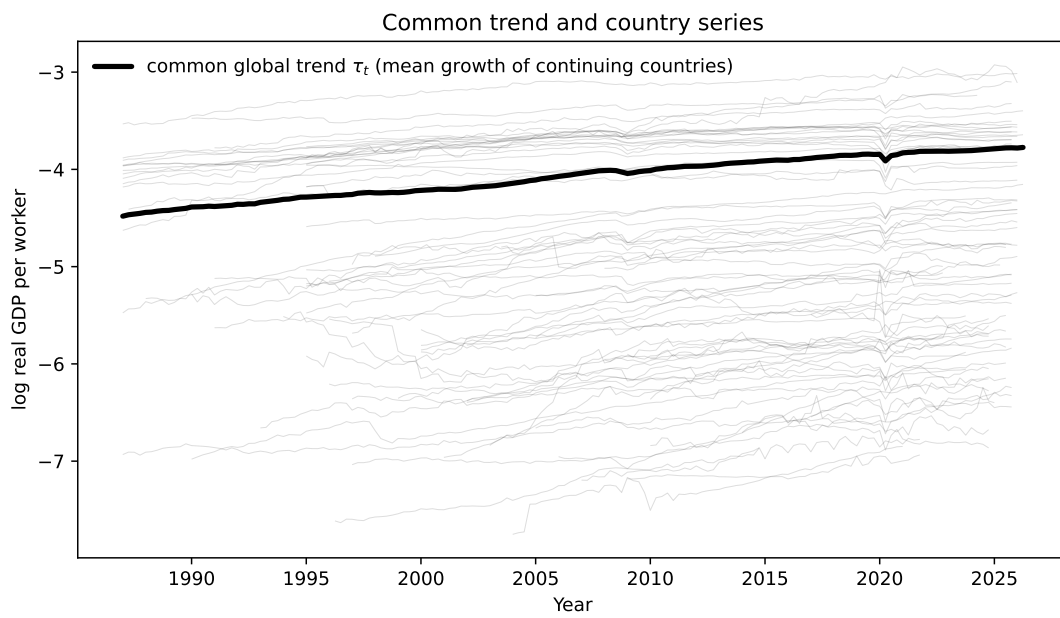


Figure 1: Faint lines: country log real GDP per worker. Thick line: common τ_t from average growth of continuing countries (new entrants do not shift the level).

2 Common RW trend removed; RE intercepts plus Barro catch-up

2026-09-17 SA panel, $y_{it} - \hat{\tau}_t$, from 1987. 74 countries, 8291 observations, 1987-Q1–2026-Q2. $|\rho| < 0.99$, $\lambda = 0.98$ fixed. Fitted countries: 74. Observations: 8291. Log-likelihood: 22680.54. Ergodic mean of the cycle $E[z] = -0.0157$ (0.0235). Permanent RE intercepts and catch-up $y_{it} = a_i + b_i \lambda^{t-T_{i0}} + z_{it}$ with $\alpha = -0.8548$ (0.0135), $\omega = 0.6826$ (0.0085), $\lambda = 0.9800$ (fixed), $\omega_b = 0.4885$ (0.0144). Posterior-mean a_i : mean -1.065, min -3.415, max 0.761. Posterior-mean b_i : mean -0.013, min -1.152, max 1.156.

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.5800 (0.1765)	0.0000 —	0.0755 (0.0416)
σ	0.0728 (0.0030)	0.0072 (0.0001)	0.0188 (0.0005)
ρ	0.9899 (0.0010)	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.6964 (0.0368)	0.0056 (0.0116)	0.2980 (0.0399)
Π_1	0.0065 (0.0026)	0.9480 (0.0052)	0.0456 (0.0057)
Π_2	0.0506 (0.0069)	0.0595 (0.0074)	0.8899 (0.0104)
π	0.0813 (0.0097)	0.4942 (0.0263)	0.4245 (0.0227)

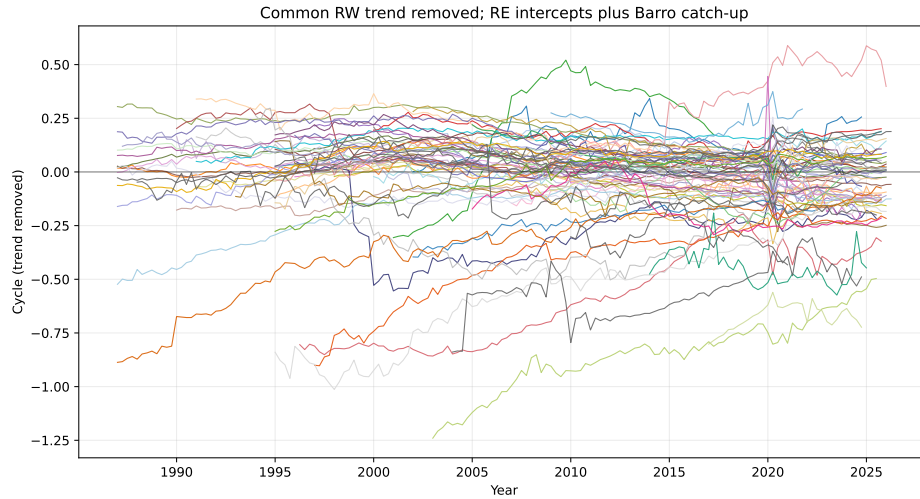


Figure 2: Country cycles after removing τ_t , a_i , and $b_i \lambda^{t-T_{i0}}$.